

LLMs as Social Network Generators

How model choice, culture, prompt language, and generation method reshape synthetic friendship networks

50

fixed personas

4

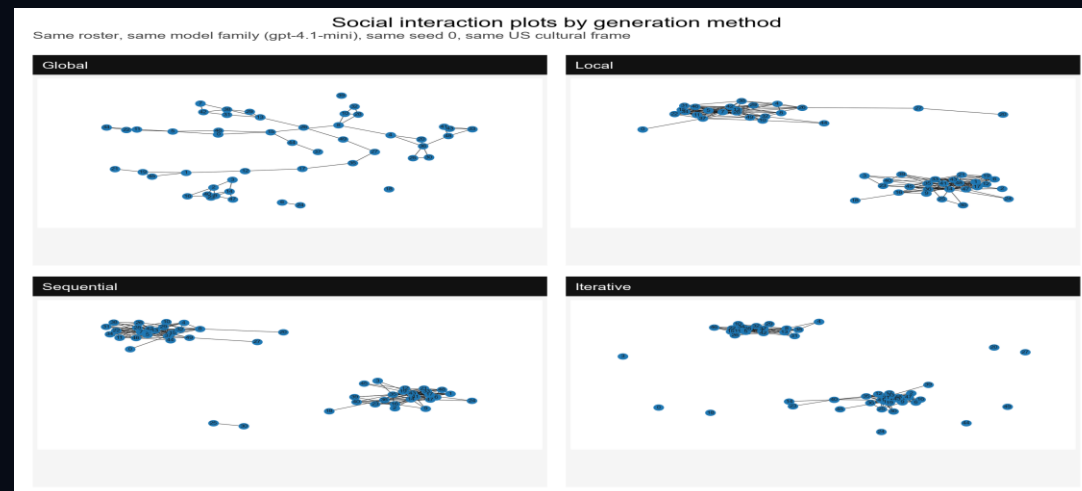
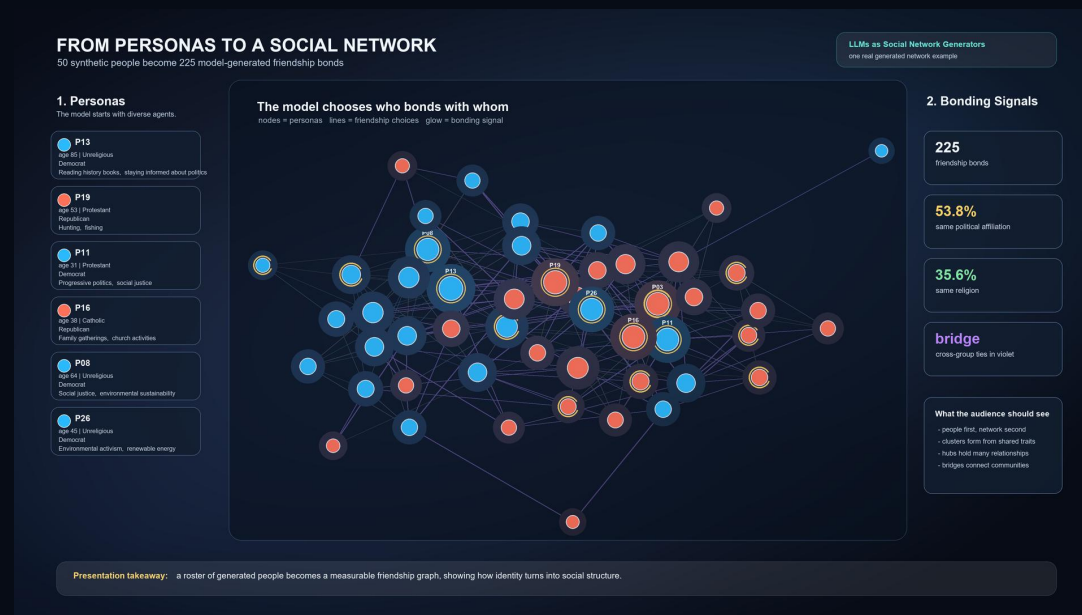
generation methods

3 + older

new GPT-4.1 family plus
base-model context

Core claim

LLMs can generate plausible social networks, but the final graph is shaped by prompt context, model choice, and generation procedure.



The Story In One Minute

From older model baselines to our controlled capstone extension.

Base repo

LLMs generated friendship graphs and compared homophily/topology against real networks.

Older outputs

Sequential baselines exist for GPT-3.5, GPT-4o, Llama 3.1, and Gemma2 variants.

Our extension

We reused one fixed 50-person roster and varied culture, method, model, and prompt language.

Final finding

Generated networks are structurally sensitive: political homophily dominates often, but culture, language, method, and model shift the result.

Why This Matters

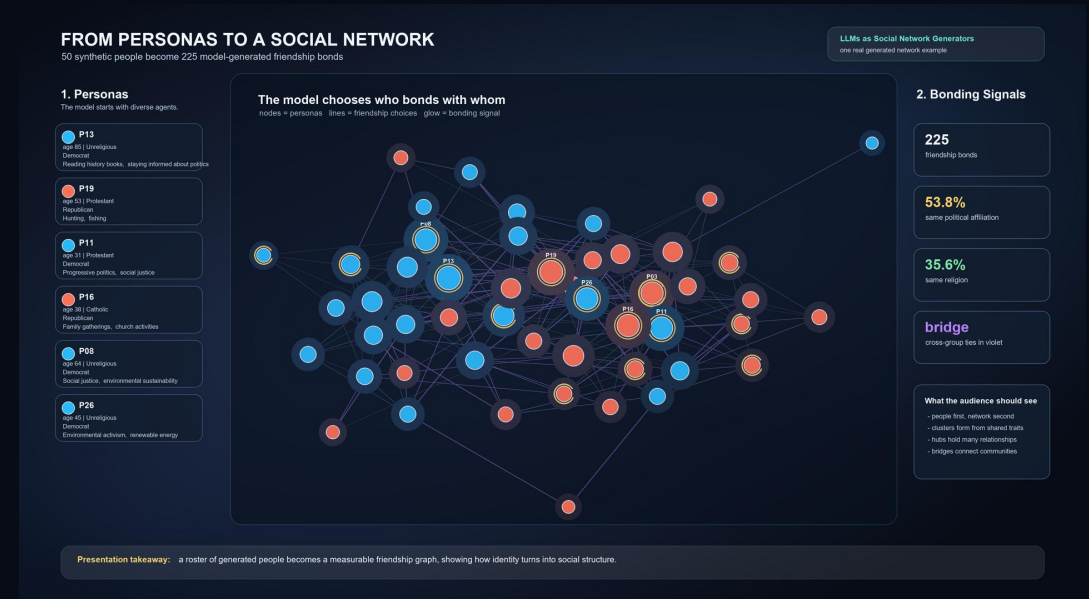
Synthetic social networks are useful only if we understand their hidden assumptions.

LLM-generated agents are increasingly used for simulations, games, education, and social-science experiments.

But a network is more than a list of personas: the model decides who connects to whom.

Small prompt/model changes can shift homophily, density, connected components, and community structure.

So the research question is about reliability: when do generated social networks stay stable, and when do they change?



Presentation phrase

"The important output is not just 50 personas. It is the relationship structure the model builds between them."

What Was Already In The Base Repository

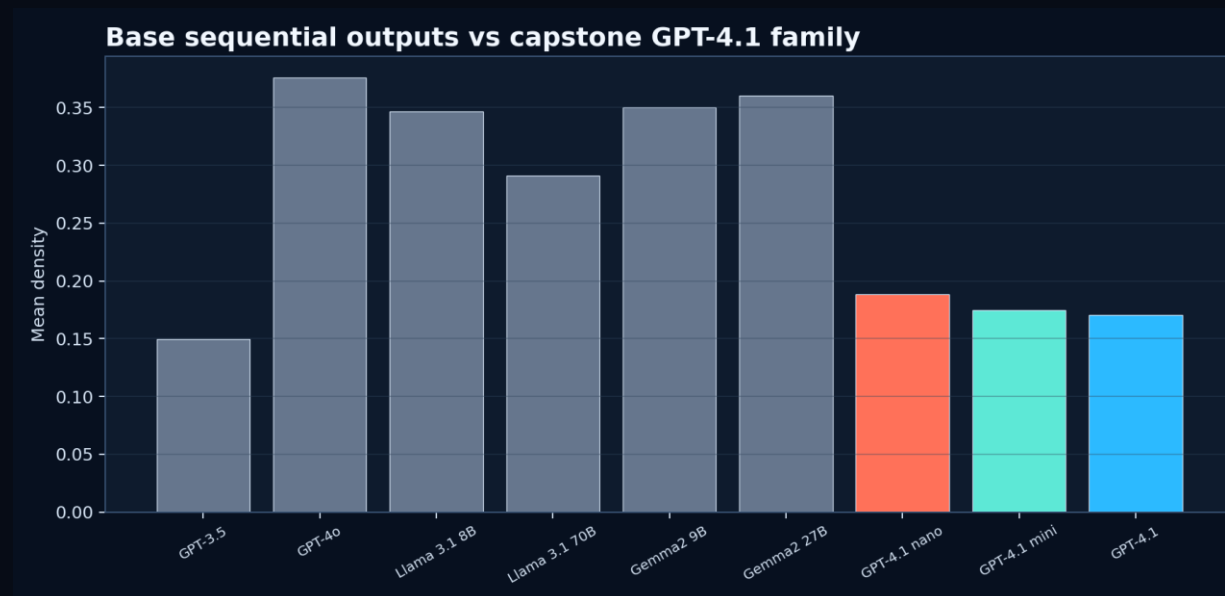
Older work established the network-generation problem and produced baseline outputs.

Base / older model side

Original framing: LLMs generate social networks and can overestimate political homophily.

Existing sequential outputs: GPT-3.5, GPT-4o, Llama 3.1 8B/70B, and Gemma2 9B/27B.

Those outputs give comparison context, but not the full capstone matrix.



How our project differs

We converted the repo into a controlled study: same personas, multiple cultures, all four methods, multiple GPT-4.1 model sizes, and fixed-culture language experiments.

Our Contribution

What we added to the older repository.

Step 1

Made the repo runnable with OpenAI-only setup and repaired path/plotting issues.

Step 2

Added cultural context: US, India, Japan, Brazil with language fixed to English.

Step 3

Expanded from sequential to four methods: sequential, global, local, iterative.

Step 4

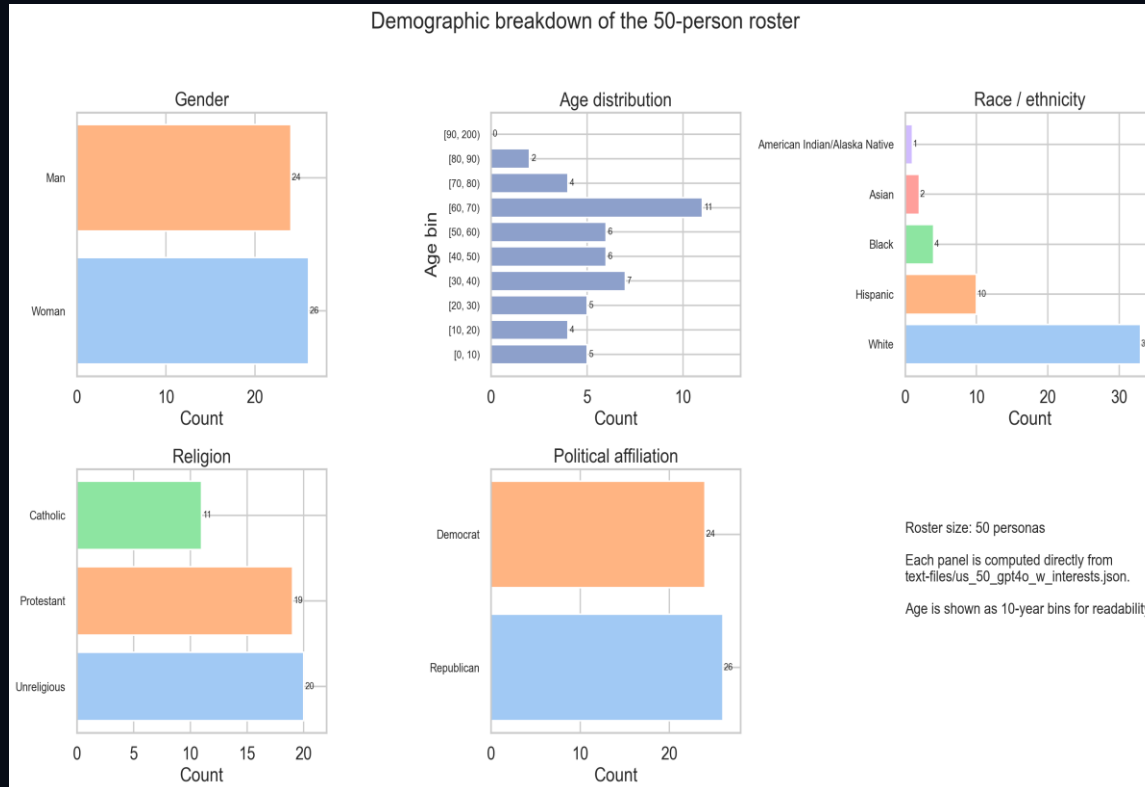
Added prompt-language study: English, Spanish, Hindi, Japanese with culture fixed to US.

Verification

Checked generated graph files, PNGs, homophily, metrics, node counts, and edge sanity.

Dataset: The Same 50 Personas Everywhere

Using one fixed roster isolates the experimental variable.



Roster file: text-files/us_50_gpt4o_w_interests.json

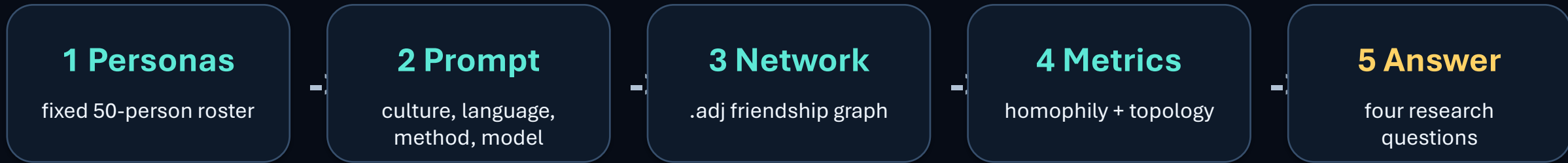
Each persona includes gender, age, race/ethnicity, religion, political affiliation, and interests.

The same 50 personas are reused across all new studies.

That means differences are caused by prompt context, model, method, or language - not by changing the people.

Pipeline

How the project turns people into measurable networks.



Homophily: do people connect more often to similar people than expected?

Topology: density, clustering, connected component size, modularity, path length.

Model divergence: edge distance between generated graphs under matched conditions.

Experiment Matrix

The capstone study is controlled but broad.

4

cultures: US, India, Japan,
Brazil

4

methods: sequential,
global, local, iterative

3

GPT-4.1 model sizes

4

prompt languages

192

verified generated graphs in
Steps 2-4

Verification passed

Step 2 cultural study: 24/24 graph
conditions passed

Step 3 method expansion: 72/72 graph
conditions passed

Step 4 language study: 96/96 graph
conditions passed

Design logic

RQ1-RQ3 combine sequential baseline
with the method expansion.

RQ4 fixes culture to US, then varies
prompt language.

All studies reuse the same 50-person
roster.

The Four Generation Methods

Different prompting procedures create different graph structures.

Sequential

People choose friends one at a time as the network grows.

Global

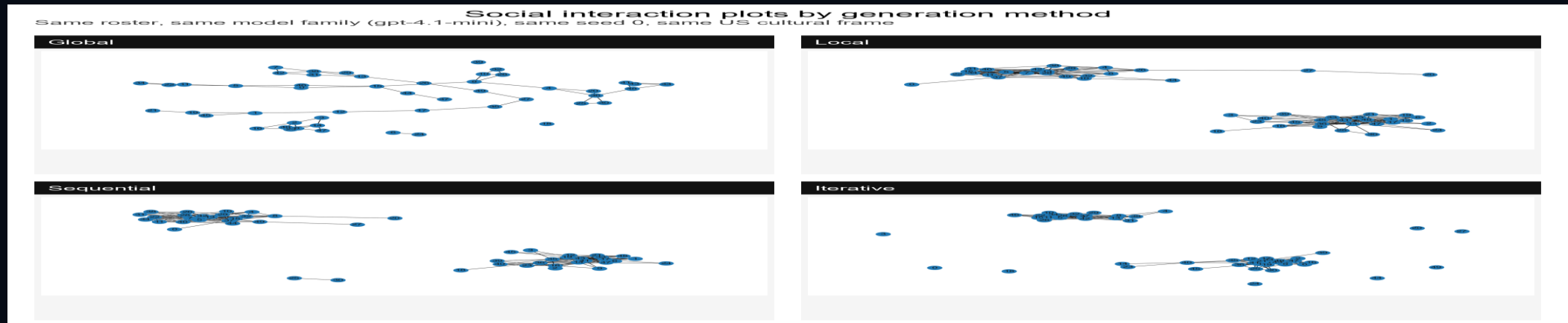
The model proposes friendship pairs for the whole network at once.

Local

Each focal person chooses friends from candidates without full sequential buildup.

Iterative

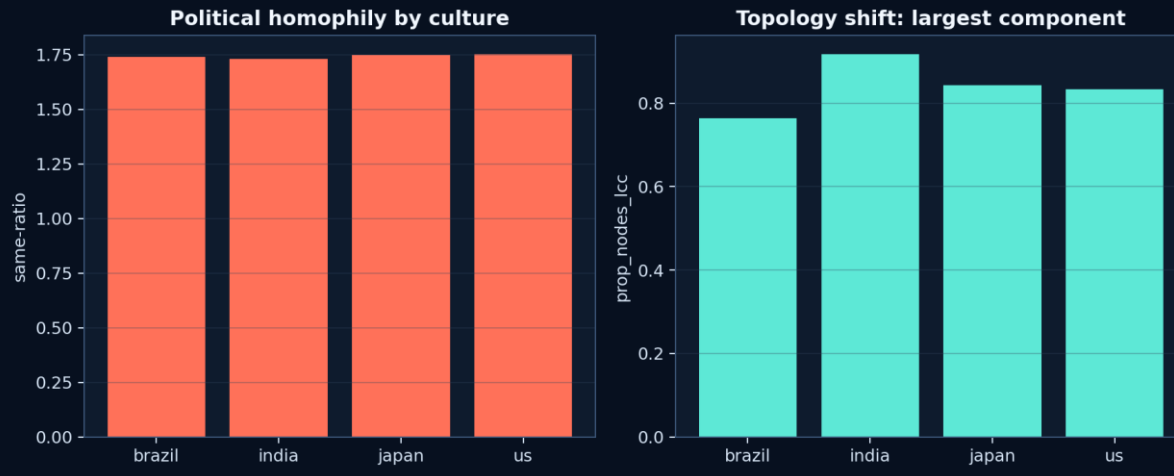
The network is revised through add/drop style friendship updates.



RQ1: Does Cultural Context Matter?

Yes. Holding language fixed to English, culture still changed homophily and topology.

RQ1: culture changes both homophily and network structure



Answer

Largest culture-driven homophily shift: political affiliation, same-ratio range 0.874.

Largest topology shift: proportion of nodes in the largest connected component, range 0.500.

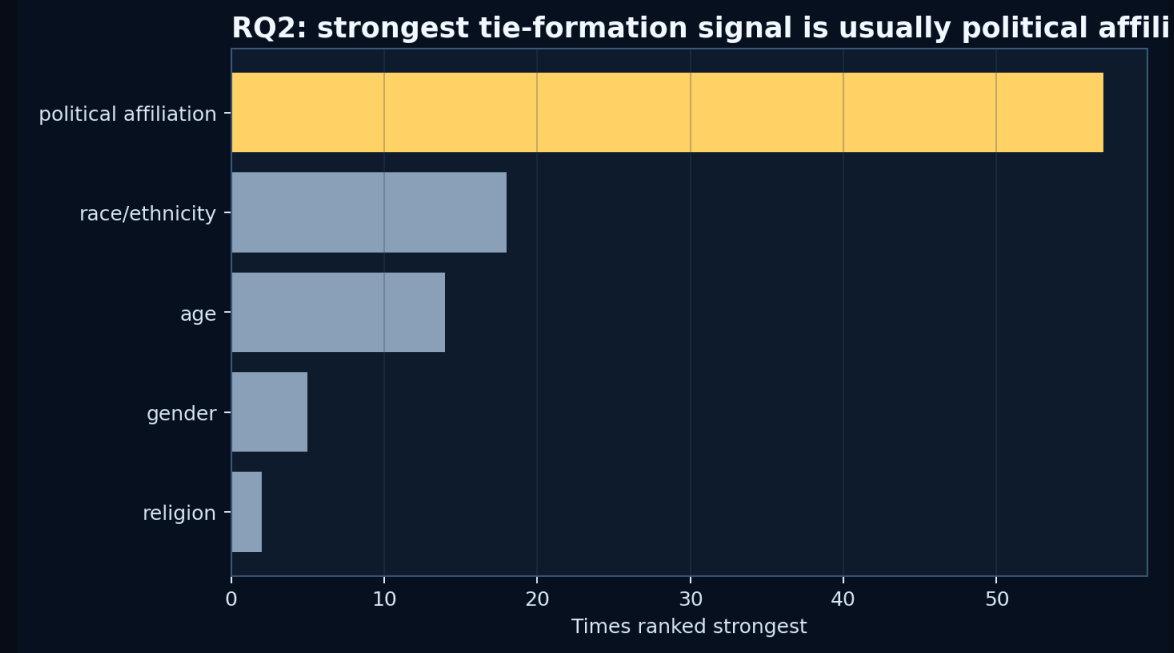
Plain meaning: changing only the cultural frame changed who clustered with whom and how connected the graph became.

Presenter line

"Culture was not cosmetic. It changed the generated social structure even when the language stayed English."

RQ2: Which Demographics Dominate Tie Formation?

Political affiliation is the strongest recurring signal, with an important global-method exception.



Across the English-language cultural study, political affiliation was the most frequent top-ranked homophily dimension.

Across the fixed-culture language study, political affiliation remained the most frequent strongest dimension.

After adding all methods, the result became more nuanced: global often elevated age, while local, sequential, and iterative more often elevated political affiliation.

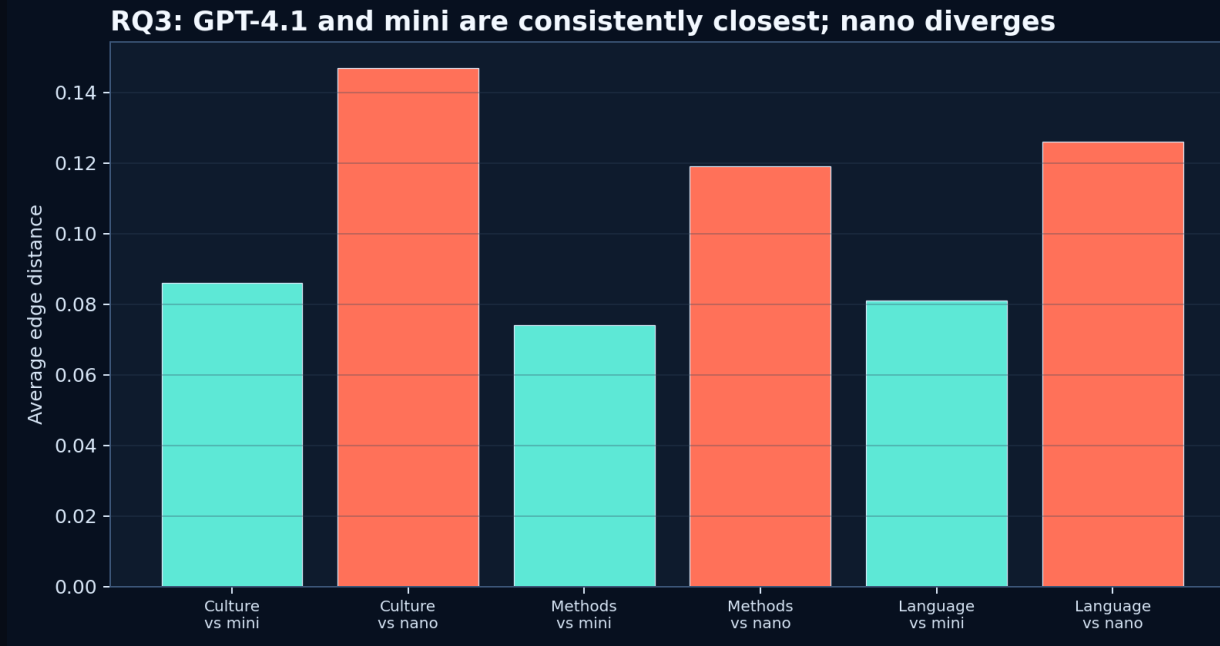
Plain-language

answer

Usually, the model formed friendships around political similarity more than the other demographic dimensions, but the prompting method can change which signal rises to the top.

RQ3: Are Models Interchangeable?

No. Model size changed the generated edge set in a repeatable way.



Closest pair in cultural study: GPT-4.1 vs GPT-4.1-mini, edge distance 0.086.

Closest pair in method study: GPT-4.1 vs GPT-4.1-mini, edge distance 0.074.

Closest pair in language study: GPT-4.1 vs GPT-4.1-mini, edge distance 0.081.

Farthest pair in all three: GPT-4.1 vs GPT-4.1-nano.

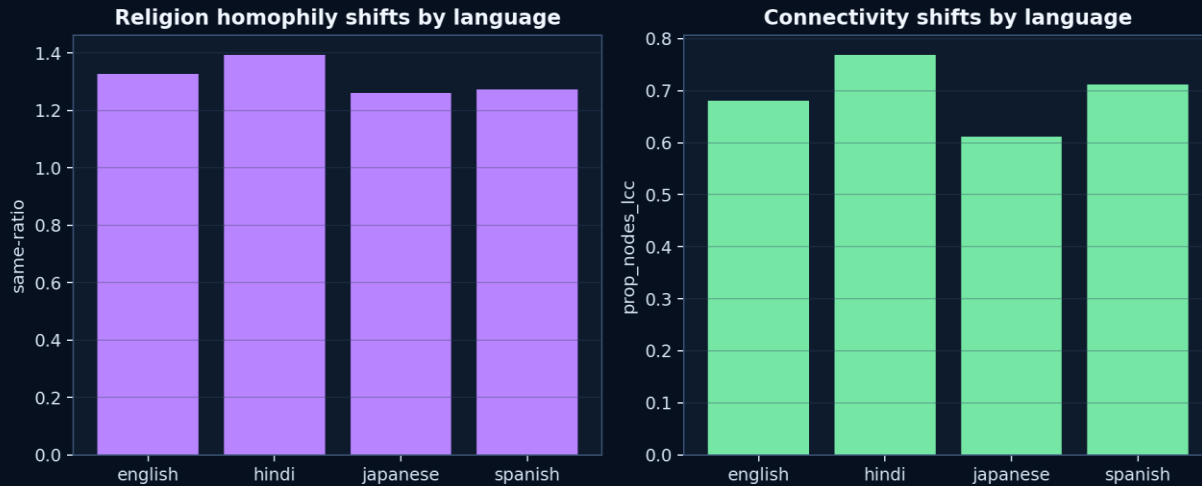
Interpretation

Nano is the outlier. GPT-4.1 and GPT-4.1-mini stay closest across culture, method, and language studies.

RQ4: Does Prompt Language Matter?

Yes. With culture fixed to US, language still changed homophily and topology.

RQ4: changing prompt language changes the generated network



Largest language-driven homophily shift: religion, range 1.502.

Largest topology shift: prop_nodes_lcc, range 0.740.

Closest language pair: Hindi vs Japanese, edge distance 0.079.

Farthest language pair: Japanese vs Spanish, edge distance 0.088.

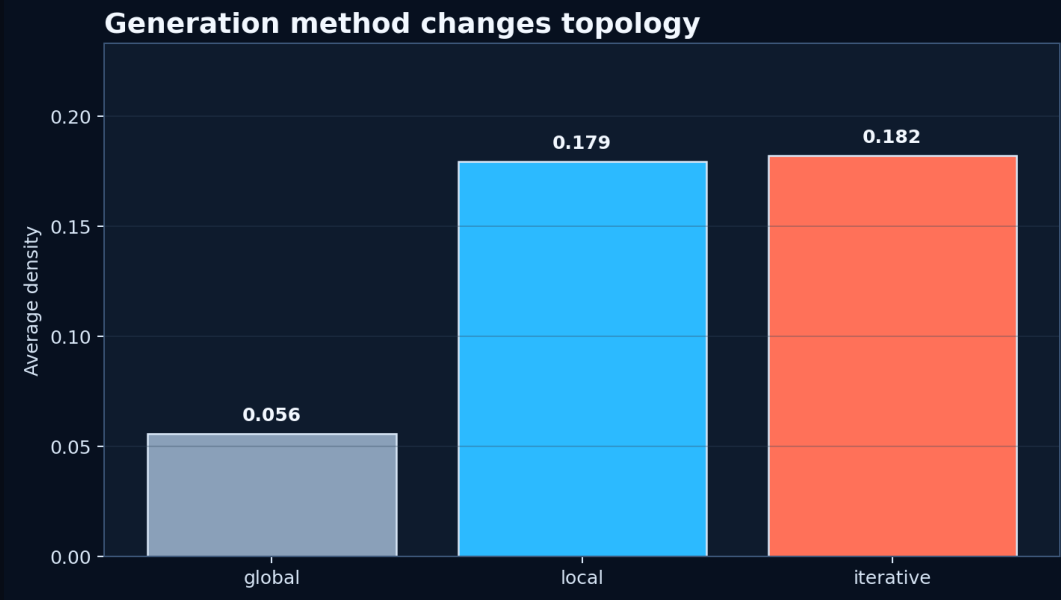
Plain-language

answer

Even when the people and culture stayed fixed, changing the language of the prompt changed the network that came out.

Method Effects

The generation procedure changes network density and connectedness.



Global is sparsest: average density 0.056.

Iterative is densest: average density 0.182.

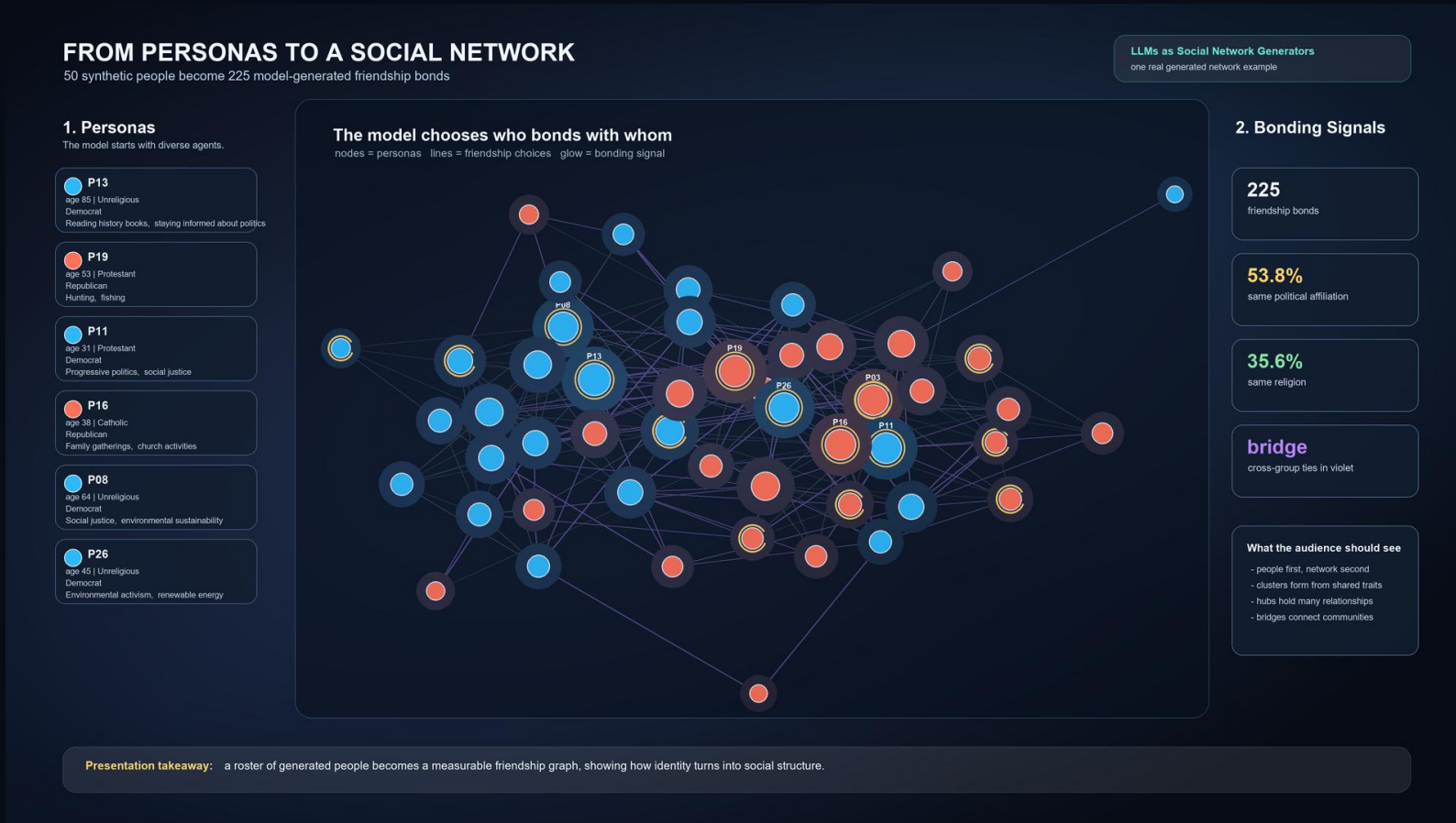
Local is close to iterative: average density 0.179.

This means methods are not neutral wrappers around the same model; the asking procedure shapes the graph.



Live Demo Slide: Personas Become Bonds

Use this when presenting the visual animation.



Say this

First, the model receives 50 synthetic personas.

Then it chooses friendship bonds.

The final graph contains hubs, clusters, and bridge people.

So the project studies generated social interaction, not only generated identities.

Optional: play the MP4 animation after this slide.

Final Answers To The Research Questions

The shortest version to say out loud.

RQ1 Culture

Yes: culture changes both homophily and topology.

RQ2 Demographics

Political affiliation is usually the strongest tie-formation signal; global often elevates age.

RQ3 Models

The models are not interchangeable. GPT-4.1 and mini are closest; nano diverges most.

RQ4 Language

Yes: even with culture fixed, prompt language changes homophily and graph structure.

Limitations And Why They Matter

Shows you understand what the results do and do not prove.

These are synthetic networks, not real friendship networks.

The fixed 50-person roster is useful for control, but it is still one roster.

Two seeds per condition give repeatability evidence, but more seeds would strengthen estimates.

Prompt wording and model provider behavior can change over time.

The findings are best interpreted as model behavior under controlled prompts, not as claims about real human societies.

Strong closing

sentence
The value of this project is that it measures those sensitivities instead of hiding them.

Conclusion

Generated networks are powerful, but they are not neutral.

Main takeaway

LLMs can create structurally plausible social networks, but culture, language, method, and model choice reshape who becomes connected to whom.

Base repo: established the network-generation problem with older model outputs.

Our work: extended it into a controlled capstone matrix.

Final result: the same personas can produce meaningfully different social structures depending on the experimental condition.

Q&A